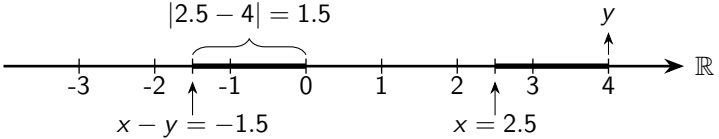
ECON 2050 – Mathematical Economics

Basic Real Analysis I

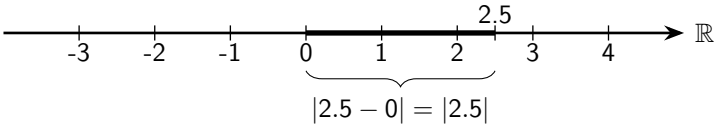
The space we work in

This course assumes basic familiarity of the set of real numbers, or the *real line*, denoted by $\mathbb{R}$.

- The **distance** between $x \in \mathbb{R}$ and $y \in \mathbb{R}$ is the *absolute value* of $x - y$, that is, $|x - y|$. If this distance is small, then x and y are "close" to each other.

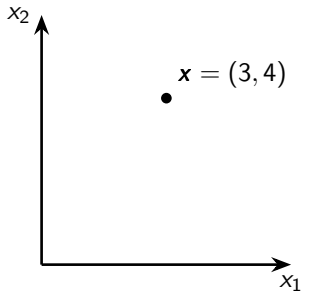


- In particular, the distance between $x \in \mathbb{R}$ and 0, $|x| = |x - 0|$, measures the "length" of x.

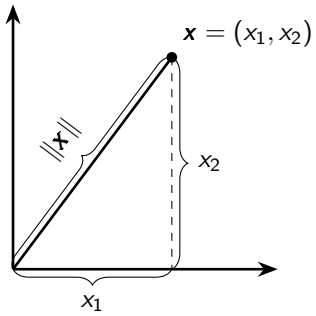


More generally, we will work with the set of all m-tuples of real numbers, denoted by $\mathbb{R}^m$, where m is a positive integer.

- A point in $\mathbb{R}^m$ can be written as $\boldsymbol{x} = (x_1, \ldots, x_m)$, where x_i is its i-th coordinate.
- When $m = 1$, we get the familiar real line $\mathbb{R}$.



We also generalize the ideas of distance and length to $\mathbb{R}^m$ with $m > 1$: as in $\mathbb{R}$, in $\mathbb{R}^m$ the **norm** (generalizing the absolute value) of $\boldsymbol{x} = (x_1, \ldots, x_m) \in \mathbb{R}^m$ is the "length" of $\boldsymbol{x}$, or the distance between $\boldsymbol{x}$ and $\mathbf{0}$; denote it by $\|\boldsymbol{x}\|$.



By Pythagoras' Theorem, $\|\boldsymbol{x}\|^2 = x_1^2 + x_2^2$, or

$$\|\boldsymbol{x}\| = \sqrt{x_1^2 + x_2^2}$$

More generally (for arbitrary positive integer m):

Definition

The **norm** of $\boldsymbol{x} = (x_1, \ldots, x_m) \in \mathbb{R}^m$ is

$$\|\boldsymbol{x}\| = \sqrt{x_1^2 + x_2^2 + ... + x_m^2}$$

The **distance** between $\boldsymbol{x} \in \mathbb{R}^m$ and $\boldsymbol{y} \in \mathbb{R}^m$ is

$$\begin{aligned}\|\boldsymbol{x} - \boldsymbol{y}\| &= \|(x_1 - y_1, x_2 - y_2, \ldots, x_m - y_m)\| \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \ldots + (x_m - y_m)^2}\end{aligned}$$

- $\mathbb{R}^m$ is sometimes called an Euclidean space.
- This definition generalizes our earlier definitions:
  - The norm of $x_1 \in \mathbb{R}$ $(= \mathbb{R}^1)$ is $\|x_1\| = \sqrt{x_1^2} = |x_1|$
  - The norm of a point $\boldsymbol{x} = (x_1, x_2) \in \mathbb{R}^2$ in a plane is $\|\boldsymbol{x}\| = \sqrt{x_1^2 + x_2^2}$

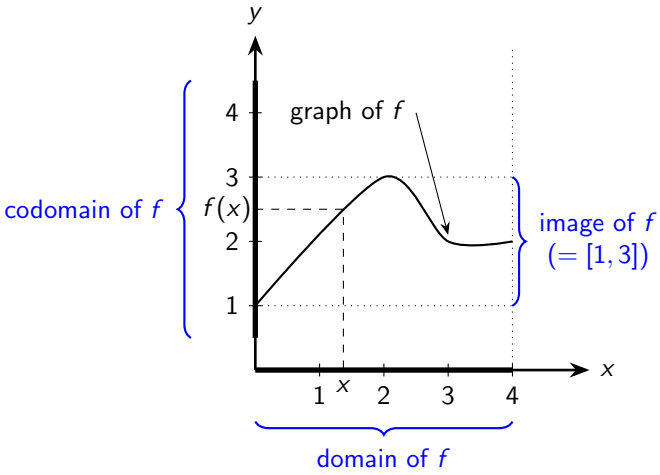
Functions (SB p. 273)

Definition

A function f from a set X to a set Y is a rule that assigns to *each* element of X *exactly one* element of Y. We write $f : X \to Y$ or $f : X \to Y, x \mapsto f(x)$.

- the **domain** of f is the set X
- the **codomain** (or **target space** in SB) of f is the set Y
- the **image** of x under f, written $f(x)$, is the element $y \in Y$ which f assigns to $x \in X$
- the **range** or **image** of f is the set $f(X) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$
- the **graph** of f is the set $\{(x, y) \in X \times Y : y = f(x)\}$

Example: $X = [0, 4]$, $Y = [0.5, 4.5]$, $f : X \to Y$ as shown below



Sequences (SB Section 12.1 & 12.2)

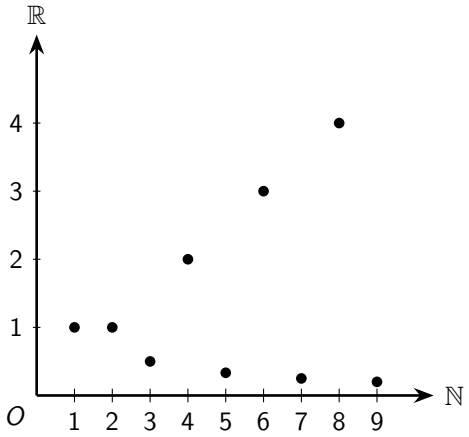
Definition

A sequence (in X) is a function $f : \mathbb{N} \to X$.

- $\mathbb{N} = \{1, 2, 3, \ldots\}$ = set of natural numbers
- we are interested in the cases $X = \mathbb{R}$ and $X = \mathbb{R}^m$ where $m \in \mathbb{N}$
- it is common to write (x_n) or $(x_n)_{n\in\mathbb{N}}$ or $(x_n)_{n=1}^{\infty}$ or $(x_1, x_2, x_3, \ldots)$ for f, where $x_n \equiv f(n)$

  Example: $(\frac{1}{n})_{n\in\mathbb{N}}$ denotes $f : \mathbb{N} \to \mathbb{R}, n \mapsto \frac{1}{n}$
- SB write $\{x_n\}$ or $\{x_n\}_{n\in\mathbb{N}}$ or $\{x_n\}_{n=1}^{\infty}$ or $\{x_1, x_2, x_3, \ldots\}$ for f

Example: Below, we illustrate the first **elements** of the sequence $f : \mathbb{N} \to \mathbb{R}$ given by $(\frac{1}{1}, 1, \frac{1}{2}, 2, \frac{1}{3}, 3, \ldots)$



Often, we want to figure out where a sequence "tends to" as the index $n \in \mathbb{N}$ becomes "large" (i.e., goes to $+\infty$)

Example: Interest rate = 3.5% (both for saving and borrowing). Your bank offers to pay you (and your offspring) \$1000/year for n years. How much should you maximally pay for this?

- If $n = 1$, then the maximum you should pay is:

$$PV_1 = 1000$$

- If $n = 2$, then the maximum you should pay is:

$$PV_2 = 1000 + \frac{1000}{1.035} \approx 1000 + 966 = 1966$$

- More generally,

$$\begin{aligned} PV_n &= 1000 + \frac{1000}{1.035^1} + \ldots + \frac{1000}{1.035^{n-1}} \\ &= \sum_{k=0}^{n-1} \frac{1000}{1.035^k} = 1000 \cdot \sum_{k=0}^{k-1} \left(\frac{1}{1.035} \right)^k \end{aligned}$$

- This defines a sequence $(PV_n)_{n=1}^{\infty}$; up to rounding:

$$(PV_n)_{n=1}^{\infty} = (1000, 1966, 2900, 3802, 4673, \ldots)$$

- Where does $(PV_n)_{n=1}^{\infty}$ "tend to" as $n \to \infty$?

$$PV_n = 1000 \quad + \frac{1000}{1.035^1} + \ldots\ldots + \frac{1000}{1.035^{n-2}} + \frac{1000}{1.035^{n-1}} \tag{1}$$

$$\frac{1}{1.035} PV_n = \frac{1000}{1.035^1} + \frac{1000}{1.035^2} + \ldots\ldots + \frac{1000}{1.035^{n-1}} + \frac{1000}{1.035^n} \tag{2}$$

$$(1) - (2): \quad \left(1 - \frac{1}{1.035}\right) PV_n = 1000 - \frac{1000}{1.035^n}. \quad \text{Therefore:}$$

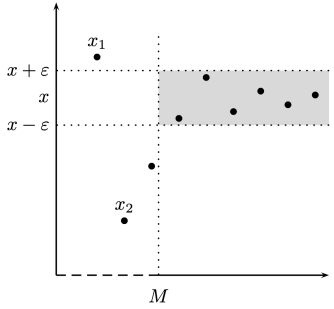
$$PV_n = \frac{1000 - \frac{1000}{1.035^n}}{1 - \frac{1}{1.035}} \approx 29571 - \frac{\frac{1000}{1.035^n}}{1 - \frac{1}{1.035}} (\approx 29571 \text{ for } n \text{ large})$$

Definition

Let $(x_n)_{n\in\mathbb{N}}$ be a sequence with codomain $\mathbb{R}$. We say that (x_n) **converges** to x^* or that x^* is the **limit** of $(x_n)_{n\in\mathbb{N}}$, and write $\lim_{n\to\infty} x_n = x^*$ or $x_n \to x^*$, if

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : |x_n - x^*| < \varepsilon.$$

If $(x_n)_{n\in\mathbb{N}}$ does not converge to any $x^* \in \mathbb{R}$ we say that $(x_n)_{n\in\mathbb{N}}$ **diverges**.



(x_m) converges to x.

Example

Consider the sequence $(\frac{1}{n})_{n\in\mathbb{N}}$, we also have $\lim_{n\to\infty}\frac{1}{n}=0$

n	(x_n)
1	1
2	0.5
3	0.333333
4	0.25
5	0.2
6	0.166667
7	0.142857
8	0.125
9	0.111111
10	0.1
11	0.090909
12	0.083333
13	0.076923

- Suppose $\epsilon = 0.1$, what is the appropriate N?
- Intuitively, we can see that when $N = 11$, all other $n > 11$, the distance between x_n and x^* is less than ϵ
- But how do we generalise this for any $\epsilon > 0$?

The idea of convergence is that the *tail* of a sequence approximates its limit to *any desired degree of accuracy*.

Some initial finitely many terms of the sequence may be quite apart from its limit, but eventually all terms of the sequence accumulates around this limit.

- To see this, consider a sequence (x_n) defined as

$$x_n = \begin{cases} n & \text{if } n \leq 100; \\ \frac{1}{n-100} & \text{if } n > 100. \end{cases}$$

- To show that $\lim_{n\to\infty} x_n = 0$, we need to pick N large enough; in fact, we need to pick $N > (1/\varepsilon) + 100$.
- So although real sequences $(1/n)$ and x_n appear to be quite different in their first 100 terms, they have the same long-run behavior: they both converge to 0.
- That is, the initial terms of the sequence have no say on the behavior of the tail of the sequence.

Sequences in $\mathbb{R}^m$

Sometimes, we want to consider sequences with codomain $\mathbb{R}^m$.

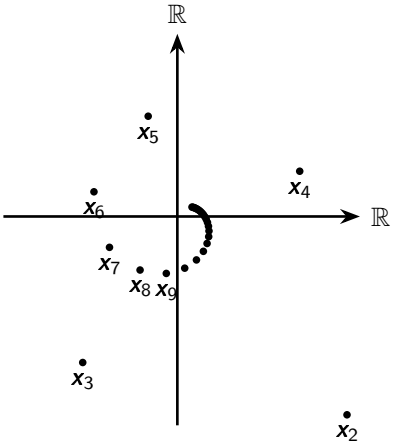
Example: Every second, we observe Ann's x_1 and x_2 coordinates (her position on a plane). At time t, her position is

$$\boldsymbol{x}_t = (x_{1,t}, x_{2,t}) = \left(\frac{30}{t}\sin\left(\frac{30}{t}\right), \frac{30}{t}\cos\left(\frac{30}{t}\right)\right)$$

Will Ann's position "converge" to some location? That is, will the sequence $(\boldsymbol{x}_t)_{t\in\mathbb{N}}$ converge?

Convention: We use $\boldsymbol{x} \in \mathbb{R}^m$ and $(x_1, \ldots, x_m) \in \mathbb{R}^m$ interchangeably.

Below a picture of a few elements of $(\boldsymbol{x}_t)_{t\in\mathbb{N}}$:



It appears as if $\boldsymbol{x}_t \to (0,0)$: Ann gets closer and closer to $(0,0)$. To verify this, let's formalize the notion of convergence in $\mathbb{R}^m$.

Definition

Let $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ be a sequence with codomain $\mathbb{R}^m$. We say that $(\boldsymbol{x}_n)$ **converges** to $\boldsymbol{x}^*$ or that $\boldsymbol{x}^*$ is the **limit** of $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$, and write $\lim_{n\to\infty} \boldsymbol{x}_n = \boldsymbol{x}^*$ or $\boldsymbol{x}_n \to \boldsymbol{x}^*$, if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : \|\boldsymbol{x}_n - \boldsymbol{x}^*\| < \varepsilon.$$

If $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ does not converge to any $\boldsymbol{x}^* \in \mathbb{R}^m$ we say that $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ **diverges**.

We can rewrite this definition using the notion of open balls:

Definition

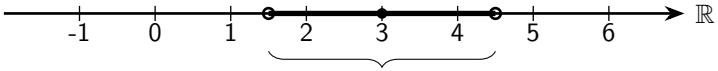
The **open ball** $B_r(\boldsymbol{x})$ with radius $r > 0$ and center $\boldsymbol{x} \in \mathbb{R}^m$ is the set of points $\boldsymbol{y} \in \mathbb{R}^m$ with distance less than r from $\boldsymbol{x}$,

$$B_r(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^m : \|\boldsymbol{y} - \boldsymbol{x}\| < r\}.$$

A sequence $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ with codomain $\mathbb{R}^m$ converges to $\boldsymbol{x}^*$ if and only if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N : \boldsymbol{x}_n \in B_\varepsilon(\boldsymbol{x}^*).$$

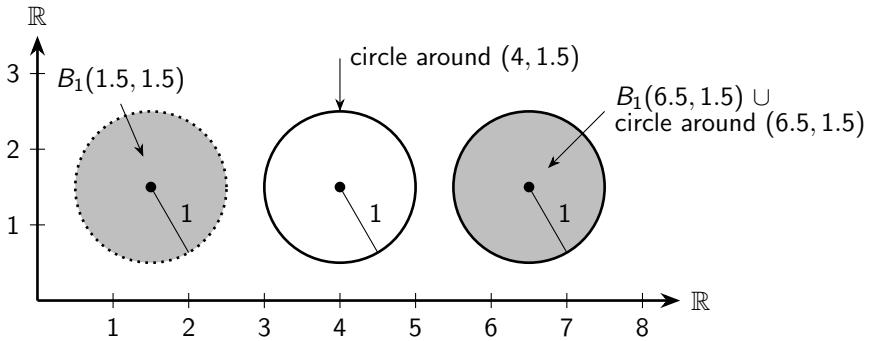
In $\mathbb{R}$ ($m = 1$), an open ball is an open interval.



$$B_{1.5}(3) = \{\, y \in \mathbb{R} : |y - 3| < 1.5\} = \{\, y \in \mathbb{R} : 1.5 < y < 4.5\} = (1.5, 4.5)$$

Note that the empty circles at the endpoints of the interval $(1.5, 4.5)$ indicate that 1.5 and 4.5 are *not* in $B_{1.5}(3)$.)

- In $\mathbb{R}^2$ ($m = 2$), the open ball $B_r(x_1, x_2)$ consists of all points in the circle with radius r around (x_1, x_2), excluding the circle's border.
- *Convention: When we draw sets, solid lines are part of the set, dotted lines are not.*



Theorem (compare SB Thm 12.1)

A sequence can have at most one limit.

Intuition: a sequence cannot get arbitrarily close and *stay* arbitrarily close to two different points.

Proof: See supplementary materials on Blackboard (ie: Logic.pdf)

Theorem

If $\lim_{n\to\infty} \boldsymbol{x}_n = \boldsymbol{x}^* \in \mathbb{R}^m$ *and* $\lim_{n\to\infty} \boldsymbol{y}_n = \boldsymbol{y}^* \in \mathbb{R}^m$ *and* $c \in \mathbb{R}$*, then*

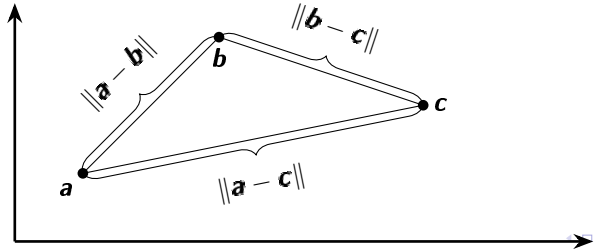
(a) $\lim_{n\to\infty}(\boldsymbol{x}_n + \boldsymbol{y}_n) = \boldsymbol{x}^* + \boldsymbol{y}^*$.

(b) $\lim_{n\to\infty} c\boldsymbol{x}_n = c\boldsymbol{x}^*$.

We will only prove part (a) and leave part (b) as an exercise. To that end, recall for $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{R}^m$:

$$\|\boldsymbol{a} - \boldsymbol{c}\| \leq \|\boldsymbol{a} - \boldsymbol{b}\| + \|\boldsymbol{b} - \boldsymbol{c}\|$$

(triangle inequality — see SB Thm 10.5).



Proof of part (a): Let $\varepsilon > 0$.

1. Since $\boldsymbol{x}_n \to \boldsymbol{x}^*$, $\exists N_1 \in \mathbb{N}$ such that $\|\boldsymbol{x}_n - \boldsymbol{x}^*\| < \varepsilon/2$ for all $n \geq N_1$.
2. Since $\boldsymbol{y}_n \to \boldsymbol{y}^*$, $\exists N_2 \in \mathbb{N}$ such that $\|\boldsymbol{y}_n - \boldsymbol{y}^*\| < \varepsilon/2$ for all $n \geq N_2$.

Therefore $\|\boldsymbol{x}_n - \boldsymbol{x}^*\| < \varepsilon/2$ and $\|\boldsymbol{y}_n - \boldsymbol{y}^*\| < \varepsilon/2$ for all $n \geq N \equiv \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$\begin{aligned}
\|(\boldsymbol{x}_n + \boldsymbol{y}_n) - (\boldsymbol{x}^* + \boldsymbol{y}^*)\| &= \|(\boldsymbol{x}_n - \boldsymbol{x}^*) - (\boldsymbol{y}^* - \boldsymbol{y}_n)\| \\
&\leq \|(\boldsymbol{x}_n - \boldsymbol{x}^*) - \boldsymbol{0}\| + \|\boldsymbol{0} - (\boldsymbol{y}^* - \boldsymbol{y}_n)\| \text{ (by triangle inequality)} \\
&= \|\boldsymbol{x}_n - \boldsymbol{x}^*\| + \|\boldsymbol{y}_n - \boldsymbol{y}^*\| \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon,
\end{aligned}$$

where $\boldsymbol{0} = (0, \ldots, 0)$. Thus, $(\boldsymbol{x}_n + \boldsymbol{y}_n) \to (\boldsymbol{x}^* + \boldsymbol{y}^*)$.

Theorem

Let $(x_n)_{n\in\mathbb{N}}$ *and* $(y_n)_{n\in\mathbb{N}}$ *be sequences with codomain* $\mathbb{R}$. *If* $\lim_{n\to\infty} x_n = x^* \in \mathbb{R}$ *and* $\lim_{n\to\infty} y_n = y^* \in \mathbb{R}$, *then*

(a) $\lim_{n\to\infty}(x_n \cdot y_n) = x^* \cdot y^*$

(b) *if there exists* $b \in \mathbb{R}$ *such that* $x_n \leq b$ *for all* $n \in \mathbb{N}$, *then* $x^* \leq b$.

Proof of part (a) is left as an exercise. See SB Thm 12.3 if you are interested.

Proof of part (b): Suppose, toward a contradiction, that $x_n \leq b$ for all $n \in \mathbb{N}$, but $x^* > b$.

- Choose ε so that $0 < \varepsilon < x^* - b$ (e.g., we can let $\varepsilon = (x^* - b)/2$). Then, $b < x^* - \varepsilon$.
- Because $x_n \to x^*$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in B_\varepsilon(x^*) = (x^* - \varepsilon, x^* + \varepsilon)$.
- This implies that $x_n \geq b$ for all $n \geq N$, a contradiction.

Theorem (compare SB Thm 12.5)

Let $(\boldsymbol{x}_n)_{n\in\mathbb{N}}$ be a sequence with codomain $\mathbb{R}^m$. Then

$$\lim_{n\to\infty} \boldsymbol{x}_n = \boldsymbol{x}^* \quad \textit{if and only if} \quad \left(\lim_{n\to\infty} x_{i,n} = x_i^* \textit{ for all } i = 1,\ldots,m\right).$$

We can use this theorem to verify that "Ann converges to $(0,0)$":

- We have
$$\lim_{t\to\infty} x_{1,t} = \lim_{t\to\infty} \frac{30}{t}\sin\left(\frac{30}{t}\right) = 0$$
- We have
$$\lim_{t\to\infty} x_{2,t} = \lim_{t\to\infty} \frac{30}{t}\cos\left(\frac{30}{t}\right) = 0$$
- Therefore,
$$\lim_{t\to\infty} \boldsymbol{x}_t = \left(\frac{30}{t}\sin\left(\frac{30}{t}\right), \frac{30}{t}\cos\left(\frac{30}{t}\right)\right) = (0,0)$$

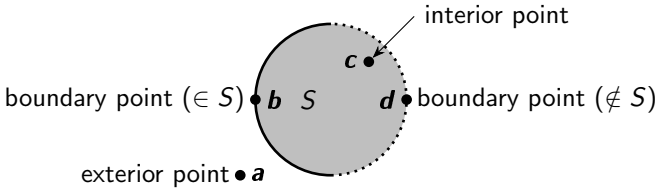
Open and closed sets (SB Section 12.3 & 12.4)

Definition

Consider a set $S \subseteq \mathbb{R}^m$. A point $\boldsymbol{x} = (x_1, \ldots, x_m) \in \mathbb{R}^m$ is

- an **interior point** of S if there exists an open ball centered at $\boldsymbol{x}$ contained entirely in S.
- an **exterior point** of S if there is an open ball centered at $\boldsymbol{x}$ located entirely outside of S.
- a **boundary point** of S if each open ball centered at $\boldsymbol{x}$ contains both a point in S and a point outside of S.

Note that a boundary point of S does not have to belong to S.



These notions allow us to define two important "topological" properties of sets:

Definition

A set $S \subseteq \mathbb{R}^m$ is

- **open** if all of its points are interior points.
- **closed** if it contains all of its boundary points.

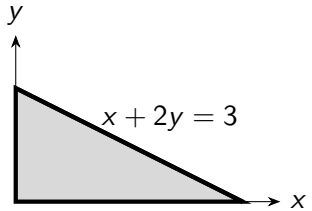
Examples:

- The set $S_1 = \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + (y-1)^2 < 1\}$ is open .
- More generally, open intervals in $\mathbb{R}$ and open balls in $\mathbb{R}^m$ are open (SB Thm 12.7).
- The set $S_2 = \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + (y-1)^2 \leq 1\}$ is closed . Its boundary is formed by the points of the set

$$S_3 = \{(x, y) \in \mathbb{R}^2 : (x-1)^2 + (y-1)^2 = 1\} \subseteq S_2.$$

- Any closed interval $[a, b] \equiv \{x \in \mathbb{R} : a \leq x \leq b\}$ is closed , with boundary points a, b.

- The set $S_3 = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0, x + 2y \leq 3\}$ is closed .

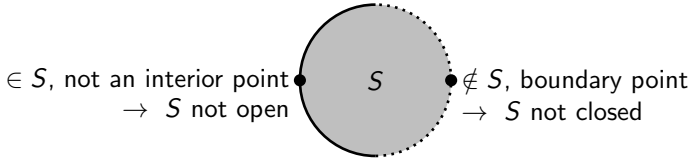


The **interior** of a set $S \subseteq \mathbb{R}^m$, denoted by $\text{int}(S)$, is the set of all interior points of S.

- A set $S \subseteq \mathbb{R}^m$ is open if and only if $S = \text{int}(S)$.

Note that some sets are **neither open nor closed.**

Example:



Bounded and compact sets (SB Section 12.5)

Besides open and closed sets, compact sets play an important role:

Definition

A set $S \subseteq \mathbb{R}^m$ is

- **bounded** if

$$\exists K > 0 \text{ such that } \forall \boldsymbol{x} \in S : \|\boldsymbol{x}\| \leq K$$

  (i.e. if there is an open ball centered at $\mathbf{0}$ that contains S).
- **compact** if it is closed and bounded.

Examples:

- Any closed interval $[a, b]$ is compact.
- Let $p_1, p_2 \geq 0$ be two prices and $w \geq 0$ be a wealth level.
  - $B_1 = \{(x_1, x_2) \in \mathbb{R}^2 : p_1 x_1 + p_2 x_2 \leq w\}$ is closed but not bounded.
  - The budget set $B_2 = \{(x_1, x_2) \in \mathbb{R}^2 : p_1 x_1 + p_2 x_2 \leq w, x_1 \geq 0, x_2 \geq 0\}$ is closed and bounded, and therefore compact.